Dividing Polynomials

Name _____

Algebra 2 Lesson 3-4 (fast track) Date _____ Period _____

Learning goal: I can divide a polynomial by a linear binomial with synthetic division and use the remainder to tell whether the binomial is a factor.

Language goal: I can explain whether a binomial is a factor of a polynomial using the words divisor, quotient, remainder and factor.

On the quiz: You will (1) divide by $x+3$ with long division (full Savvas quiz only), (2) finish a synthetic division by $x-2$ and read the quotient, (3) find the remainder for $x+2$ two ways and see they match (bonus), (4) find a remainder when a term is missing (full Savvas quiz only), and (5) use a remainder of 0 to write $P(x)$ as a product of two factors.

On the Topic 3 test: Question 1 asks for a remainder (Example 2). Question 5 asks for $P(x)$ as a product of factors (Example 5). Question 10 asks you to finish a synthetic division (Example 2).

Example 2: Use Synthetic Division to Divide by $x - a$

What is $(2x^3 - 7x^2 - 4)$ divided by $(x - 3)$? Use synthetic division.

The idea: synthetic division divides a polynomial by a linear expression $x - a$ (leading coefficient 1). You write only a, the **zero of the divisor**, and the coefficients of the dividend.

Step 1. The divisor $x-3$ is 0 when $x = 3$, so $a = 3$. Write the dividend as $2x^3 - 7x^2 + 0x - 4$: the coefficients are 2, -7, 0, -4. (Use 0 for the missing x term.)

Step 2. Bring down the first coefficient, 2. Multiply a by it: $3 \cdot 2 = 6$. Add to the next coefficient: $-7 + 6 = -1$.

Step 3. Repeat until every column is done:
$3 \cdot (-1) = -3$, $0 + (-3) = -3$;
$3 \cdot (-3) = -9$, $-4 + (-9) = -13$.

Step 4. Read the last row: 2, -1, -3 are the coefficients of the **quotient**, one degree lower than the dividend (cubic ÷ linear = quadratic). The last number, -13, is the **remainder**.

Step 2

3	2	-7	0	-4
		6		
	2	-1		

Steps 3 and 4

3	2	-7	0	-4
		6	-3	-9
	2	-1	-3	**-13**
	quotient coefficients			remainder

Answer: $\dfrac{2x^3 - 7x^2 - 4}{x - 3} = 2x^2 - x - 3 - \dfrac{13}{x - 3}$, so $2x^3 - 7x^2 - 4 = (2x^2 - x - 3)(x - 3) - 13$.

Check (quotient × divisor + remainder): $(2x^2 - x - 3)(x - 3) = 2x^3 - 7x^2 + 9$; $9 - 13 = -4$. ✓

Synthetic division works only when the divisor is $x - a$; for $x^2 + 5$ or $2x + 3$, use long division.

Say it: To divide by $x - 3$ I use $a =$ _____. The last row gives the _____ $2x^2 - x - 3$ and the _____ -13. The _____ $x - 3$ is not a factor, because ...

Try It 2: Use synthetic division

Use synthetic division to divide $(3x^3 - 5x + 10)$ by $(x - 1)$.

$a =$ _____ Is a term missing? _____
Coefficients: _____

Quotient: _____

Remainder: _____

$\dfrac{3x^3 - 5x + 10}{x - 1} =$ _____

Check: quotient × divisor + remainder:

Remainder Theorem and Factor Theorem

The **Remainder Theorem** states that if a polynomial $P(x)$ is divided by $x - a$, the remainder is $P(a)$.

The **Factor Theorem** states that the expression $x - a$ is a factor of a polynomial $P(x)$ if and only if $P(a) = 0$.

Why: dividing gives $P(x) = Q(x)(x - a) + r$. Put in $x = a$: $P(a) = Q(a) \cdot 0 + r = r$. So the remainder is 0 exactly when $P(a) = 0$.

Example 5: Check Whether $x - a$ Is a Factor of $P(x)$

How can you use the Factor Theorem to determine whether the given binomial is a factor of $P(x)$? If it is a factor, write the polynomial in factored form.

A. $P(x) = x^4 - 8x^3 + 16x^2 - 23x - 6$; binomial: $x - 6$. The binomial $x - 6$ is a factor of $P(x)$ if 6 is a zero of $P(x)$.

Method 1 Synthetic substitution ($a = 6$).

6	1	−8	16	−23	−6
		6	−12	24	6
	1	−2	4	1	**0**

Method 2 Direct substitution.

$$\begin{aligned} P(6) &= 6^4 - 8(6^3) + 16(6^2) - 23(6) - 6 \\ &= 1{,}296 - 1{,}728 + 576 - 138 - 6 \\ &= 0 \end{aligned}$$

Because $P(6) = 0$, the Factor Theorem says $x - 6$ **is** a factor. The last row gives the other factor:
$P(x) = (x^3 - 2x^2 + 4x + 1)(x - 6)$.

B. $P(x) = x^5 - 5x^3 + 9x^2 - x + 3$; binomial: $x + 3$. Write $x + 3 = x - (-3)$, so $a = -3$. There is no x^4 term: use 0.

Method 1 Synthetic substitution ($a = -3$).

-3	1	0	-5	9	-1	3
		-3	9	-12	9	-24
	1	-3	4	-3	8	**-21**

Method 2 Direct substitution.

$$\begin{aligned} P(-3) &= (-3)^5 - 5(-3)^3 + 9(-3)^2 - (-3) + 3 \\ &= -243 + 135 + 81 + 3 + 3 \\ &= -21 \end{aligned}$$

Because $P(-3) = -21 \neq 0$, -3 is not a zero of $P(x)$, so $x + 3$ is **not** a factor of $P(x)$. Both methods give the same number, -21: that is the Remainder Theorem.

Say it: $x - 6$ is a _____ of $P(x)$ because the _____ is 0. $x + 3$ is not a factor because $P($_____$) =$ _____, and ...

Try It 5: Is the binomial a factor?

Use the Factor Theorem to determine whether the given binomial is a factor of $P(x)$.

a. $P(x) = x^3 - 10x^2 + 28x - 16$;
binomial: $x - 4$
$a =$ _____

Remainder: _____ Factor? _____
If it is a factor, $P(x) =$

b. $P(x) = 2x^4 + 9x^3 - 2x^2 + 6x - 40$;
binomial: $x + 5$
$a =$ _____

Remainder: _____ Factor? _____
Check one answer with direct substitution, $P(a)$:

Explain your answer to part b. Use the words *remainder* and *factor*.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
D6D ★	Divide polynomials using synthetic division	Quiz 2: Example 2 (also Quiz 1, full Savvas quiz only, with $a = -3$)
CHC ★	Evaluate polynomials using synthetic division	Quiz 3 (bonus): remainder $= P(a)$, Example 5; Quiz 4 (full Savvas quiz only)
VXE ★	Factor theorem	Quiz 5: remainder 0 means a factor, Example 5
8GN	Multiply polynomials	Warm-up: check a division (quotient × divisor + remainder)
NCY	Factor a polynomial given a root	Optional: the Exit question; Lesson 3-5

Exit question: The DVD stand (DOK 3)

Dajuan wants to determine the length and height of his DVD stand. The function $f(x) = x^3 + 14x^2 + 57x + 72$ represents the volume of the DVD stand, where the width is $x + 3$ units. What are possible dimensions for the length and height of the DVD stand? Explain.

$a =$ _____

Quotient: _____

Factor the quotient: _____

Length and height: _____ and _____

Explain: $x + 3$ is a _____ of $f(x)$ because the _____ is _____, so ...

Before the quiz, finish these sentences.

To divide by $x + 2$ with synthetic division, I use $a =$ _____, because $x + 2 = x -$ (_____).

If a term is missing from the dividend, I write _____ as its coefficient.

When $P(x)$ is divided by $x - a$, the _____ equals $P(a)$.

The binomial $x - a$ is a _____ of $P(x)$ exactly when the remainder is _____.